

Productivity Paradox

Measured productivity gains lag the deployment of a transformative technology

GLOSSARY 05 - SHEET 1 / 8

PROFESSIONAL DEFINITION

The productivity paradox is the observed discrepancy between rapid deployment of a powerful information technology and the absence of a corresponding rise in measured aggregate productivity. The modern explanation is that general purpose technologies require large, slow and unmeasured investments in complementary intangible assets - process redesign, skills, organisational capital - which depress measured productivity while they are being made and raise it only afterwards (Solow, 1987; Brynjolfsson, Rock and Syverson, 2021).

WHO DEFINED IT ORIGINALLY

Robert M. Solow's 1987 New York Times Book Review remark that computers were visible everywhere except in the productivity statistics named the phenomenon. Erik Brynjolfsson and Lorin Hitt (1996, 2003) at MIT established the firm-level resolution; Brynjolfsson, Daniel Rock and Chad Syverson (2021) formalised the J-curve account in the American Economic Journal: Macroeconomics.

WHY THIS DEFINITION WAS DEVELOPED

The paradox had been used to argue either that the technology was overrated or that measurement was defective. The J-curve account supplies a third explanation with a testable implication: firms making the complementary intangible investment record lower measured productivity during the investment period and higher productivity afterwards, so the aggregate lag is a composition effect rather than a measurement error.

FIGURE 1 - THE PRODUCTIVITY J-CURVE

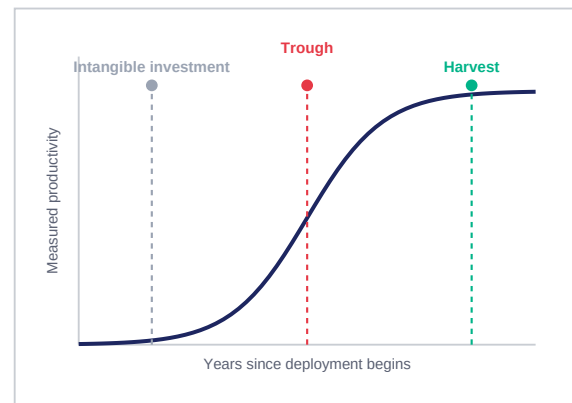


Figure 1. Measured productivity falls before it rises, because the complementary intangible investment is expensed while its output is not yet capitalised. Appraisal horizons shorter than the full curve observe only the decline (after Brynjolfsson, Rock and Syverson, 2021).

HOW IT IS USED

When appraising an AI investment, budget explicitly for the complementary intangible investment and set the measurement window beyond the trough; a two-year appraisal horizon will observe only the negative portion of the curve.

WHEN IT IS USED

At business-case appraisal for large AI programmes, in board discussion of why the technology has not yet appeared in results, at national or sectoral policy analysis, and when investment is threatened by disappointing interim returns.

BY WHOM IT IS USED

Chief financial officers setting appraisal horizons, economists and policy analysts, board members interpreting interim performance, and transformation directors defending multi-year programmes.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor account was mismeasurement: output was said to be rising in ways the national accounts did not capture. The J-curve model retains mismeasurement but relocates it to the asset side - the intangible investment is expensed rather than capitalised - which generates a specific, falsifiable time path rather than a general caution. It also yields a managerial prescription that the mismeasurement account did not: budget the complement, extend the window.

RELATED DEFINITIONS IN THIS COURSE

General Purpose Technology - Unit 5

Benefits Realisation - Unit 4

Digital Maturity - Unit 1

Skill-Biased Technological Change - Unit 5

AI Adoption - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

- Brynjolfsson, E. and Hitt, L.M. (1996) 'Paradox lost? Firm-level evidence on the returns to information systems spending', *Management Science*, 42(4), pp. 541-558.
- Brynjolfsson, E., Rock, D. and Syverson, C. (2021) 'The productivity J-curve: how intangibles complement general purpose technologies', *American Economic Journal: Macroeconomics*, 13(1), pp. 333-372.
- Solow, R.M. (1987) 'We'd better watch out', *New York Times Book Review*, 12 July, p. 36.
- Syverson, C. (2017) 'Challenges to mismeasurement explanations for the US productivity slowdown', *Journal of Economic Perspectives*, 31(2), pp. 165-186.